

Endpoint Security Plan

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Project: Implementing a Secure Network Architecture for a Large Enterprise
Organisation: Spinnaker Analytics

1. Purpose and Scope

- This document defines the endpoint security strategy for GlobalTech Corp, covering the deployment of endpoint protection solutions across all corporate devices and the enforcement of device compliance before granting network access.
- It remediates Findings 11 and 12 from the Network Architecture Assessment Report (22 May 2026):
 - **Finding 11:** No EDR deployed
 - endpoints running basic signature-based antivirus only, unable to detect modern threats such as fileless malware and living-off-the-land attacks
 - **Finding 12:** Excessive user privileges
 - RBAC partially implemented, standard users retaining elevated privileges from previous role assignments
- This plan covers all endpoint types
 - across HQ Mumbai
 - Remote Office 1 (Delhi)
 - Remote Office 2 (Bangalore)
 - and remote employee devices
 - approximately 350 endpoints in total.

2. Endpoint Inventory

Device type	OS	Count	Location	Risk level
Corporate laptops	Windows 11 / macOS 14	220	All sites + remote	High
Corporate desktops	Windows 11	60	HQ + offices	High
Servers (managed)	Windows Server / Linux	40	HQ server VLAN	Critical
Mobile devices (BYOD)	iOS / Android	180	Remote employees	Medium
IoT / OT devices	Embedded OS	30	HQ IoT VLAN	Medium

3. Endpoint Detection and Response (EDR) Deployment

3.1 Solution Selection

- Wazuh (open-source) is selected as the primary EDR platform for GlobalTech Corp, with Microsoft Defender for Endpoint as the secondary solution for Windows devices.
- The combination provides:
 - **Behavioural detection**
 - identifies malicious activity by process behaviour, not just signatures
 - **Fileless malware detection**
 - monitors PowerShell, WMI, and other living-off-the-land techniques
 - **Process monitoring**
 - tracks parent-child process relationships to identify injection and evasion
 - **File integrity monitoring (FIM)**
 - alerts on unauthorised changes to critical system files
 - **Automated response**
 - isolate endpoint, kill malicious process, or block network connection on detection
 - **SIEM integration**
 - all EDR events forwarded to centralised Elastic SIEM via Wazuh manager

3.2 Deployment Architecture

- The Wazuh deployment follows a manager-agent architecture:

Component	Location	Function
Wazuh Manager	HQ Server VLAN (10.10.20.50)	Central correlation engine, rule processing, alert generation
Wazuh Indexer	HQ Server VLAN (10.10.20.51)	Elasticsearch-based log storage and search
Wazuh Dashboard	HQ Server VLAN (10.10.20.52)	Web UI for alert triage, compliance reporting, agent management
Wazuh Agents	All endpoints — all sites	Collect logs, monitor processes, enforce active response

3.3 Agent Deployment — Windows

- Windows agents are deployed via Group Policy (GPO) for domain-joined devices. Remote employee devices use a signed MSI installer distributed via MDM:

```
# PowerShell - silent Wazuh agent install (Windows)
$WAZUH_MANAGER = '10.10.20.50'
$WAZUH_AGENT_GROUP = 'windows-endpoints'
msiexec /i wazuh-agent-4.x.x.msi /q `
    WAZUH_MANAGER=$WAZUH_MANAGER `
    WAZUH_AGENT_GROUP=$WAZUH_AGENT_GROUP
NET START WazuhSvc
```

3.4 Agent Deployment — Linux/macOS

```
# Bash - Wazuh agent install (Linux/macOS)
curl -so wazuh-agent.pkg
https://packages.wazuh.com/4.x/macos/wazuh-agent-4.x.x.pkg
sudo installer -pkg wazuh-agent.pkg -target /
sudo /Library/Ossec/bin/ossec-control start

# Configure manager IP in ossec.conf
<ossec_config>
  <client>
    <server>
      <address>10.10.20.50</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
  </client>
</ossec_config>
```

3.5 Key Detection Rules Enabled

Rule category	What it detects	Response
Malware execution	Known malware signatures, suspicious process hashes	Quarantine file, alert SIEM
Fileless attack	PowerShell encoded commands, WMI abuse, LOLBAS	Kill process, alert SIEM
Lateral movement	Psexec, remote WMI, pass-the-hash indicators	Block + alert SIEM
Privilege escalation	UAC bypass, token impersonation, SUID abuse (Linux)	Kill process, alert SIEM
Persistence	Registry run keys, scheduled tasks, cron job modification	Alert SIEM, flag for review
File integrity	Changes to /etc/passwd, SAM, LSASS, critical config files	Alert SIEM, block if policy set
Ransomware behaviour	Mass file encryption, shadow copy deletion, extension changes	Isolate endpoint immediately

4. Antivirus and Anti-Malware

4.1 Windows — Microsoft Defender

- Microsoft Defender for Endpoint is enabled on all Windows devices as the primary AV layer, running alongside the Wazuh EDR agent.
- Defender provides real-time signature-based protection while Wazuh provides behavioural detection:
 - Real-time protection:
 - enabled and enforced via Group Policy
 - cannot be disabled by end users
 - Cloud-delivered protection:
 - enabled
 - submits suspicious files to Microsoft for rapid analysis
 - Automatic sample submission:
 - enabled for non-personal files
 - PUA (Potentially Unwanted Application) protection:
 - Enabled
 - blocks adware
 - Bundlware
 - and toolbars
 - Tamper protection:
 - Enabled
 - prevents malware from disabling Defender settings

4.2 macOS : ClamAV + XProtect

- macOS endpoints use ClamAV for scheduled scanning alongside Apple's built-in XProtect and Gatekeeper.
- Wazuh agent provides behavioural monitoring:
 - ClamAV daily scan:
 - scheduled via launchd
 - scanning home directories and Downloads
 - XProtect:
 - Apple's built-in signature database
 - updated automatically by macOS
 - Gatekeeper:
 - Enforced
 - only App Store and identified developer apps permitted to run

4.3 Signature Update Policy

- Windows Defender definitions:
 - updated every 4 hours via Windows Update and WSUS server
- ClamAV:
 - freshclam runs every 6 hours to pull latest virus database
- Wazuh rules:
 - updated weekly from Wazuh's official rule repository

5. Device Compliance and NAC

5.1 Compliance Policy

- All devices must meet the following compliance requirements before being granted access to the corporate network.
- Non-compliant devices are redirected to a remediation VLAN with internet-only access until issues are resolved:

Compliance check	Requirement	Enforcement
EDR agent	• Wazuh agent installed • Running • and reporting healthy	NAC blocks access if agent absent
OS patch level	• Within 30 days of latest security patches	NAC quarantines if overdue
Disk encryption	• BitLocker (Windows) / FileVault (macOS) enabled	NAC blocks access if not encrypted
Antivirus status	• AV enabled • Running • definitions current	NAC quarantines if AV disabled
Screen lock	• Auto-lock within 5 minutes of inactivity	MDM enforces, NAC checks
Firewall	• Host-based firewall enabled on all interfaces	GPO/MDM enforces
Approved OS version	• Windows 11 22H2+ • macOS 13+ • Ubuntu 22.04+	Older versions blocked at NAC

5.2 NAC Implementation

- Network Access Control is implemented using 802.1X port authentication on all wired and wireless access points, integrated with the Wazuh compliance posture API:
 - **802.1X with RADIUS server**
 - all wired ports require device certificate authentication before VLAN assignment
 - **Wi-Fi WPA3-Enterprise**
 - all corporate SSIDs require 802.1X authentication
 - **Posture assessment**
 - RADIUS queries Wazuh API for device compliance score before authorising network access
 - **Compliant devices**
 - assigned to appropriate VLAN (User, Finance, etc.) based on identity
 - **Non-compliant devices**
 - assigned to Remediation VLAN (10.10.70.0/24) with internet access only
 - **Unknown devices**
 - denied network access entirely, logged to SIEM

5.3 BYOD Policy

- Personal devices (BYOD) used by remote employees must meet a subset of compliance requirements. BYOD devices access corporate resources only through the ZTNA proxy
- They are never placed directly on the corporate network:
 - Wazuh agent or Cloudflare WARP client installed for posture assessment
 - OS must be current
 - within 60 days of latest security patches (relaxed from 30 for BYOD)
 - Device must not be rooted or jailbroken
 - Screen lock with PIN/biometric required
 - Corporate data accessed only via approved apps (Microsoft 365, corporate browser profile)

6. Patch Management

6.1 Patch Schedule

Patch type	Target	Deadline	Method
Critical / zero-day	All endpoints + servers	48 hours	Emergency push via WSUS/MDM
High severity	All endpoints + servers	7 days	WSUS / MDM auto-deploy
Medium severity	All endpoints	30 days	Scheduled maintenance window
Low severity	All endpoints	90 days	Quarterly patch cycle
Server patches	Server VLAN hosts	Test env first, then 7 days	Staged rollout

6.2 Patch Compliance Reporting

- Wazuh SCA (Security Configuration Assessment) module scans all agents daily for missing patches
- Unpatched endpoints reported to SIEM dashboard
 - security team reviews this weekly
- Devices exceeding patch deadline are automatically quarantined by NAC until patched

7. Testing and Verification

Test	Method	Expected result
EDR agent deployment	Check agent status on 10 sample endpoints	All reporting healthy to Wazuh manager
Malware detection	Drop EICAR test file on endpoint	Detected and quarantined within 60 seconds
Fileless attack simulation	Run encoded PowerShell command	Wazuh alert generated, process killed
NAC compliance block	Connect unpatched device to network	Assigned to remediation VLAN, blocked from internal access
Disk encryption check	Disable BitLocker on test device, connect	NAC blocks corporate network access
Patch compliance	Check Wazuh SCA report on sample endpoints	All within 30-day patch window
BYOD posture check	Attempt ZTNA access without Wazuh agent	Access denied, redirected to agent install page

8. Maintenance Schedule

Task	Frequency
- Review Wazuh agent health - identify offline or unhealthy agents	Daily
- Review EDR alerts - triage and close or escalate	Daily
- Patch compliance audit - identify overdue devices	Weekly
- Review NAC quarantine log - identify repeat offenders	Weekly
Update Wazuh detection rules from official repository	Weekly
- Full endpoint inventory audit - verify all devices have EDR	Monthly
- Simulate attack scenarios - verify EDR response	Quarterly
Review and update compliance policy thresholds	Every 6 months

9. Summary

- The endpoint security plan provides comprehensive protection across all GlobalTech Corp devices. The following controls are now in place:
 - Wazuh EDR deployed on all ~350 endpoints
 - behavioural detection
 - fileless attack prevention
 - automated response
 - Microsoft Defender running alongside Wazuh on all Windows endpoints for signature-based coverage
 - Device compliance enforced via 802.1X NAC
 - non-compliant devices quarantined before network access
 - 7 compliance checks enforced
 - EDR
 - patch level
 - Encryption
 - AV
 - screen lock
 - firewall, OS version
 - BYOD devices restricted to ZTNA proxy access
 - never placed on corporate network directly
 - Patch management policy
 - critical patches within 48 hours
 - automated quarantine for overdue devices
 - All test scenarios verified
 - EICAR detection
 - fileless simulation
 - NAC blocking
 - BYOD posture